
Please answer part I and part II on separate booklets.
Each question is 1 point (total 10 points). Answers must be justified.

1 Discrimination

The Becker model of taste-based discrimination is represented by Figure 1, where b is the minority group and a is the majority group, L is the number of workers and w is the wage.

- 1.1 Why does this model predict that the “marginal” level of prejudice should matter more than the “average” level of prejudice?
- 1.2 Is taste-based discrimination compatible with a competitive environment?
- 1.3 Summarize the pros and cons of sending fictitious resumes to potential employers to measure the extent of discrimination.

2 Segregation

There are two types of households $j = H, L$ and two neighborhoods $i = a, b$ of size N . Each group of households is also of size N . Households' valuation depends on the neighborhood social environment $s_i = H_i/N$ (where H_i is the number of rich households in neighborhood i). We assume that this valuation takes the following form: $V_H(s_i) = \ln(1 + s_i)$ and $V_L(s_i) = \ln(1 + 3s_i)$ and prices are set by competitive bidding.

- 2.1 In the real world, why should valuation increase with the share of type-H households?
- 2.2 Why is full integration an equilibrium in this model?
- 2.3 Is full integration a stable equilibrium? What about full segregation?

3 Intertemporal consumption

Utility from consumption c is given by $u(c) = \ln(2c + Z)$ (with $Z \in \mathbf{R}$). People live for two periods. They have A initial endowments. Let β their discount factor and r the interest rate. The price of the consumption good is normalized to 1.

- 3.1 Does consumption in period 1 increases with r ?
 - 3.2 Imagine now that each agent also has a kid (with total utility $V_k(c)$) in period 2. Will she give her kid some money?
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4 Crime

Legal income is $\underline{Y}$, while criminal income is $\bar{Y} > \underline{Y}$. However, criminals may get caught with probability π , in which case they only receive $\bar{Y} - \xi < \underline{Y}$. Utility associated with income Y is given by $U(Y)$, with $U'(Y) > 0$.

- 4.1 Show that both π and ξ act as deterrent against crime. What do π and ξ represent in the real world?
- 4.2 Why does the magnitude of these deterrent effects depend on attitudes toward risk?
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Figure 1: Discrimination in the Becker model

